

Demographic Analysis: Japan & United States

Report

2026-03-30

Contents

1	Part 1 – Analysis of Japanese Demographic Data	2
1.1	Data Loading & Cleaning	2
1.2	1.1 Prefecture-Level Population Change (1995–2020)	2
1.3	1.2 Time-Series Line Plot – % Change Relative to 1995	4
1.4	1.3 1995 Population vs. 1995–2020 % Change	6
1.5	1.4 City Size and Population Change	8
1.6	1.5 Population Density vs. % Population Growth (Prefectures)	9
2	Part 2 – U.S. Census & TidyCensus Analysis	12
2.1	2.1 Variable Verification	12
2.2	2.2 County Population Change 2000–2020	12
2.3	2.3 Loading RUCC Codes	13
2.4	2.4 Scatter Plots – County Growth Rate vs. Total Population	14
	2.4.1 2.4a All Counties (no grouping)	14
	2.4.2 2.4b Coloured by RUCC Code	15
	2.4.3 2.4c Metro vs. Nonmetro (RUCC 1–3 vs. 4–9)	16
2.5	2.5 Population Pyramids – Illinois Counties	17
	2.5.1 Helper: Age/Sex Variables	17
	2.5.2 Illinois: Metro vs. Nonmetro	18
	2.5.3 Midwest: Metro, Micropolitan & Rural (Three-way Comparison)	20
2.6	2.6 Choropleth Map – Midwest County Population Change 2000–2020	22
3	Summary	25
3.1	Key Findings – Japan	25
3.2	Key Findings – United States	25

```

library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggrepel)
library(scales)
library(tidycensus)
library(tigris)
library(sf)
library(patchwork)
library(knitr)
library(kableExtra)
library(viridis)
library(RColorBrewer)
library(stringr)
library(purrr)

options(tigris_use_cache = TRUE)
options(scipen = 999)

# Census API key
# Replace the string below with your key, or set it once in .Renviron:
# census_api_key("YOUR_KEY", install = TRUE)
census_api_key("cae79ee00ab828454493cd615494c2a5e", overwrite = TRUE, install = FALSE)

```

1 Part 1 – Analysis of Japanese Demographic Data

1.1 Data Loading & Cleaning

```
# Subdivisions (population by year)
subdiv_raw <- read_excel("japan-subdivisions.xlsx")

# Standardise column names and drop the trailing arrow column
subdiv <- subdiv_raw %>%
  select(Name, Status, Native, pop1995, pop2000, pop2005, pop2010, pop2015, pop2020) %>%
  mutate(across(starts_with("pop"), as.numeric))

# Areas / density
areas_raw <- read_excel("japan-prefecture-areas.xlsx")

# Row 0 is an embedded sub-header - drop it and rename columns
areas <- areas_raw %>%
  slice(-1) %>%
  select(
    Rank          = 1,
    Prefecture    = 2,
    Japanese      = 3,
    Area_km2      = 4,
    Population    = 7,
    Density       = 8,
    Capital       = 12
  ) %>%
  mutate(
    Rank          = as.integer(Rank),
    Area_km2      = as.numeric(Area_km2),
    Population    = as.numeric(gsub(",", "", as.character(Population))),
    Density       = as.numeric(Density)
  )

# Quick sanity check
cat("Subdivisions:", nrow(subdiv), "rows |",
    "Unique statuses:", paste(unique(subdiv$Status), collapse = ", "), "\n")
```

```
## Subdivisions: 1166 rows | Unique statuses: Prefecture, District (-gun), City (-shi), Subprefecture (-subgo)
```

```
cat("Areas:", nrow(areas), "rows\n")
```

```
## Areas: 48 rows
```

1.2 1.1 Prefecture-Level Population Change (1995–2020)

```
# Filter to prefectures only
prefectures <- subdiv %>%
  filter(Status == "Prefecture") %>%
```

```

mutate(
  pct_2000 = (pop2000 - pop1995) / pop1995 * 100,
  pct_2005 = (pop2005 - pop1995) / pop1995 * 100,
  pct_2010 = (pop2010 - pop1995) / pop1995 * 100,
  pct_2015 = (pop2015 - pop1995) / pop1995 * 100,
  pct_2020 = (pop2020 - pop1995) / pop1995 * 100
)

# Summary table
pref_table <- prefectures %>%
  select(Name, pop1995, pct_2000, pct_2005, pct_2010, pct_2015, pct_2020) %>%
  arrange(desc(pct_2020))

kable(pref_table,
  digits = 1,
  col.names = c("Prefecture", "Pop 1995",
    "% Δ 2000", "% Δ 2005", "% Δ 2010", "% Δ 2015", "% Δ 2020"),
  caption = "Percentage population change relative to 1995 for all 47 prefectures",
  booktabs = TRUE,
  format = "latex") %>%
  kable_styling(latex_options = c("striped", "scale_down", "HOLD_position"),
    font_size = 7)

```

Table 1: Percentage population change relative to 1995 for all 47 prefectures

Prefecture	Pop 1995	% Δ 2000	% Δ 2005	% Δ 2010	% Δ 2015	% Δ 2020
Okinawa	1273440	3.5	6.9	9.4	12.6	15.2
Kanagawa	8245903	3.0	6.6	9.7	10.7	12.0
Shiga	1287005	4.3	7.3	9.6	9.8	9.8
Aichi	6868336	2.5	5.6	7.9	9.0	9.8
Saitama	6759311	2.6	4.4	6.4	7.5	8.7
Chiba	5797782	2.2	4.5	7.2	7.3	8.4
Fukuoka	4933393	1.7	2.4	2.8	3.4	4.1
Hyōgo	5401877	2.8	3.5	3.4	2.5	1.2
Ōsaka	8797268	0.1	0.2	0.8	0.5	0.5
Miyagi	2328739	1.6	1.4	0.8	0.2	-1.1
Kyōto	2629592	0.6	0.7	0.2	-0.7	-2.0
Tochigi	1984390	1.0	1.6	1.2	-0.5	-2.6
Shizuoka	3737689	0.8	1.5	0.7	-1.0	-2.8
Hiroshima	2881748	-0.1	-0.2	-0.7	-1.3	-2.8
Ibaraki	2955530	1.0	0.7	0.5	-1.3	-3.0
Okayama	1950750	0.0	0.3	-0.3	-1.5	-3.2
Gumma [Gunma]	2003540	1.1	1.0	0.2	-1.5	-3.2
Mie	1841358	0.9	1.4	0.7	-1.4	-3.9
Ishikawa	1180068	0.1	-0.5	-0.9	-2.2	-4.0
Gifu	2102442	0.3	0.2	-1.0	-3.4	-5.9
Kumamoto	1859793	0.0	-0.9	-2.3	-4.0	-6.5
Nagano	2191857	1.0	0.2	-1.8	-4.2	-6.6
Fukui	826996	0.2	-0.7	-2.5	-4.9	-7.3
Nara	1430862	0.8	-0.7	-2.1	-4.7	-7.4
Kagawa	1027006	-0.4	-1.4	-3.0	-4.9	-7.5
Toyama	1123125	-0.2	-1.0	-2.7	-5.1	-7.9
Yamanashi	881996	0.7	0.3	-2.1	-5.3	-8.2
Hokkaidō	5692321	-0.2	-1.1	-3.3	-5.5	-8.2
Saga	884316	-0.9	-2.0	-3.9	-5.8	-8.2
Ōita	1231306	-0.8	-1.8	-2.8	-5.3	-8.7
Miyazaki	1175819	-0.5	-1.9	-3.5	-6.1	-9.0
Tottori	614929	-0.3	-1.3	-4.3	-6.7	-10.0
Ehime	1506700	-0.9	-2.6	-5.0	-8.1	-11.4
Kagoshima	1794224	-0.4	-2.3	-4.9	-8.1	-11.5
Niigata	2488364	-0.5	-2.3	-4.6	-7.4	-11.5
Shimane	771441	-1.3	-3.8	-7.0	-10.0	-13.0
Tokushima	832427	-1.0	-2.7	-5.6	-9.2	-13.6
Yamaguchi	1555543	-1.8	-4.0	-6.7	-9.7	-13.7
Fukushima	2133592	-0.3	-2.0	-4.9	-10.3	-14.1
Wakayama	1080435	-1.0	-4.1	-7.2	-10.8	-14.6
Iwate	1419505	-0.2	-2.4	-6.3	-9.9	-14.7
Yamagata	1256958	-1.0	-3.2	-7.0	-10.6	-15.0
Nagasaki	1544934	-1.8	-4.3	-7.6	-10.9	-15.1
Kōchi	816704	-0.3	-2.5	-6.4	-10.8	-15.3
Aomori	1481663	-0.4	-3.0	-7.3	-11.7	-16.4
Akita	1213667	-2.0	-5.6	-10.5	-15.7	-20.9

1.3 1.2 Time-Series Line Plot – % Change Relative to 1995

```
# Pivot to long format for plotting
pref_long <- prefectures %>%
  select(Name, pct_2000, pct_2005, pct_2010, pct_2015, pct_2020) %>%
  pivot_longer(
    cols      = starts_with("pct_"),
    names_to  = "Year",
    values_to = "pct_change"
```

```

) %>%
mutate(Year = as.integer(str_extract(Year, "\\d{4}")))

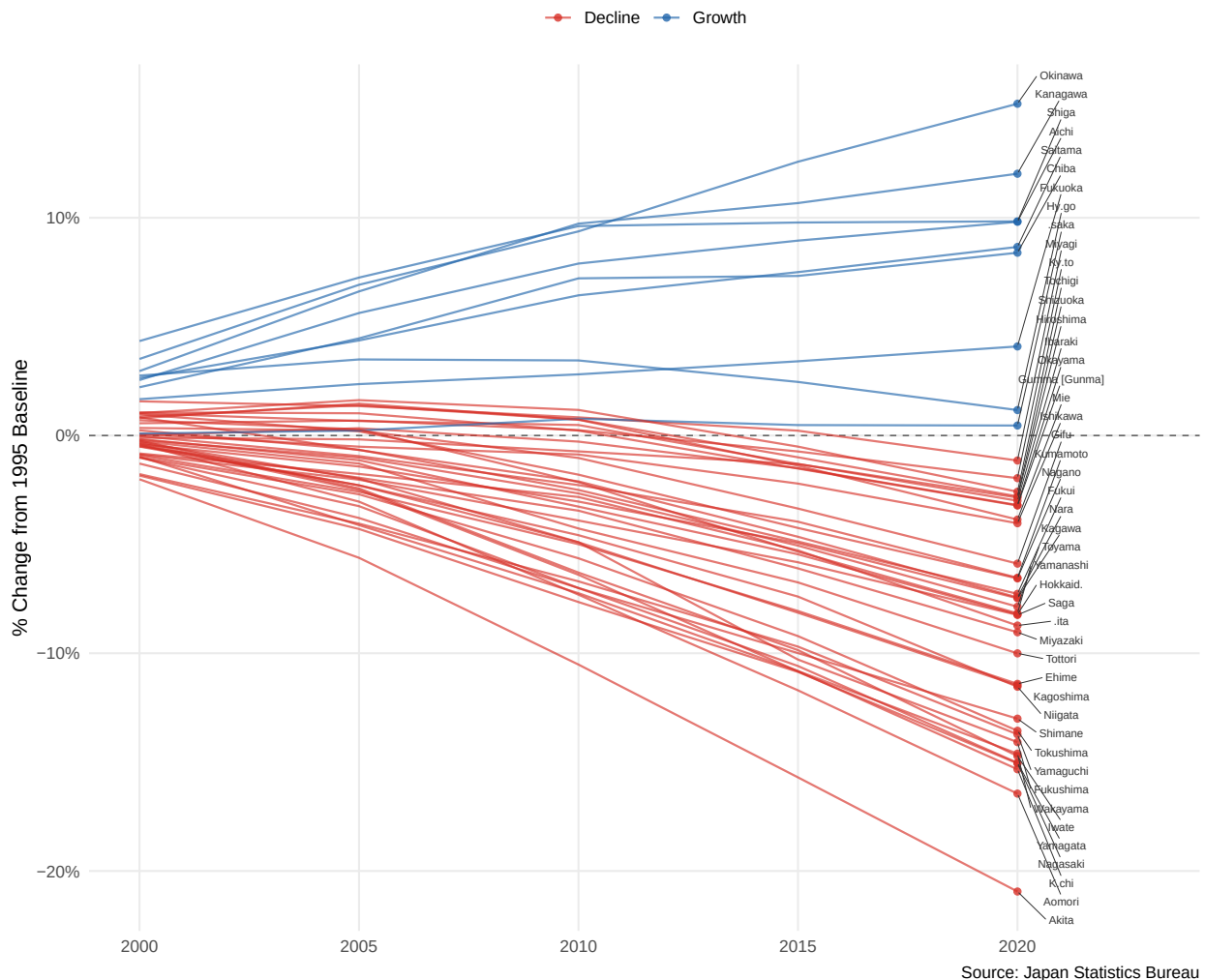
# Label only endpoints
endpoint_labels <- pref_long %>%
  filter(Year == 2020) %>%
  arrange(pct_change)

# Colour-code growth vs decline
pref_long <- pref_long %>%
  left_join(
    prefectures %>% select(Name, pct_2020),
    by = "Name"
  ) %>%
  mutate(direction = ifelse(pct_2020 >= 0, "Growth", "Decline"))

ggplot(pref_long, aes(x = Year, y = pct_change,
                      group = Name, colour = direction)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey40", linewidth = 0.4) +
  geom_line(alpha = 0.65, linewidth = 0.55) +
  geom_point(data = filter(pref_long, Year == 2020),
            size = 1.4, alpha = 0.8) +
  geom_text_repel(
    data = endpoint_labels,
    aes(label = Name),
    size = 2.2,
    direction = "y",
    nudge_x = 1,
    segment.size = 0.2,
    max.overlaps = 15,
    colour = "grey20"
  ) +
  scale_colour_manual(values = c("Growth" = "#2166ac", "Decline" = "#d73027")) +
  scale_x_continuous(breaks = c(2000, 2005, 2010, 2015, 2020),
                    limits = c(2000, 2023)) +
  scale_y_continuous(labels = label_number(suffix = "%")) +
  labs(
    title = "Percentage Population Change Relative to 1995",
    subtitle = "All 47 Japanese Prefectures · 2000-2020",
    x = NULL,
    y = "% Change from 1995 Baseline",
    colour = NULL,
    caption = "Source: Japan Statistics Bureau"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    legend.position = "top",
    panel.grid.minor = element_blank()
  )

```

Percentage Population Change Relative to 1995 All 47 Japanese Prefectures · 2000–2020



1.4 1.3 1995 Population vs. 1995–2020 % Change

```

pref_scatter <- prefectures %>%
  mutate(
    label_flag = abs(pct_2020) > 8 | pop1995 > 5e6,
    pop_m      = pop1995 / 1e6
  )

# Pearson correlation
r <- cor(pref_scatter$pop1995, pref_scatter$pct_2020, use = "complete.obs")

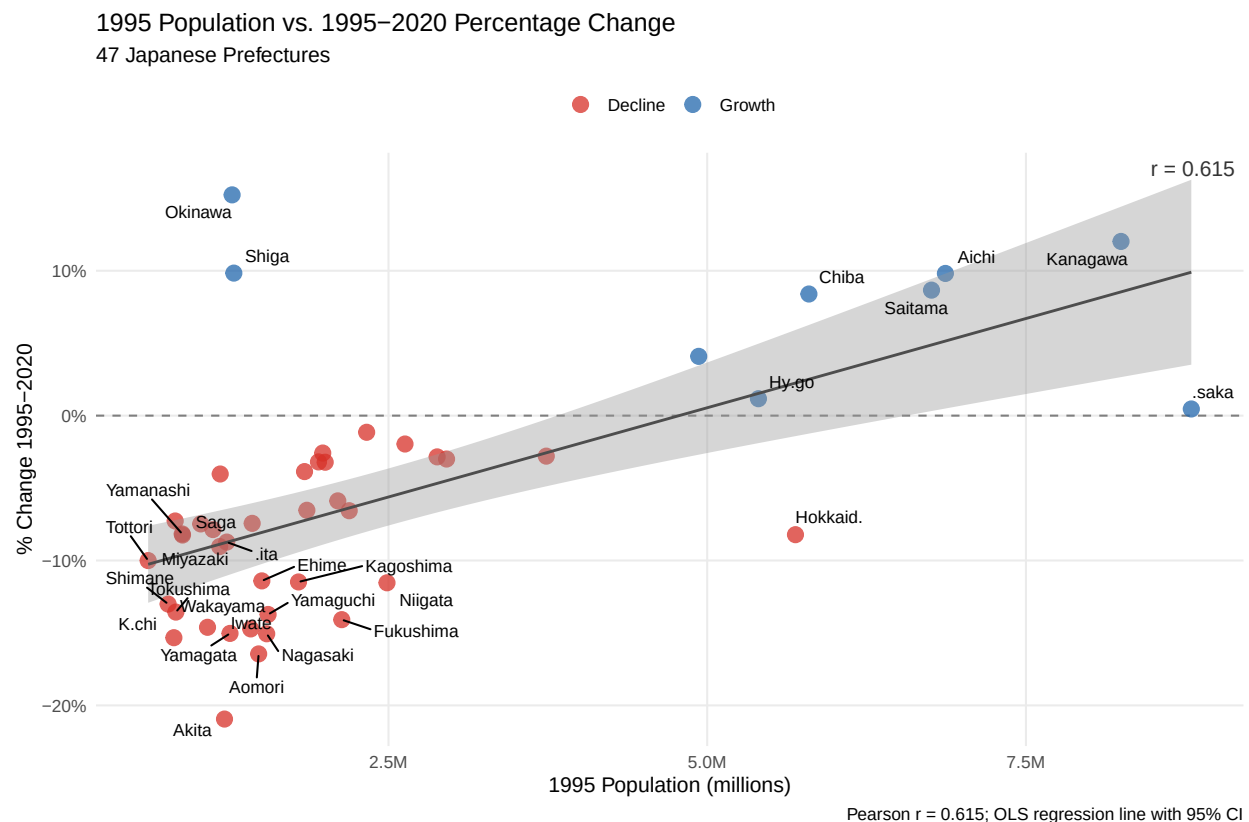
ggplot(pref_scatter, aes(x = pop_m, y = pct_2020)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(aes(colour = pct_2020 >= 0), size = 3.5, alpha = 0.75) +
  geom_smooth(method = "lm", se = TRUE, colour = "grey30",
    linewidth = 0.7, linetype = "solid") +

```

```

geom_text_repel(
  data      = filter(pref_scatter, label_flag),
  aes(label = Name),
  size      = 3,
  box.padding = 0.35,
  max.overlaps = 15
) +
scale_colour_manual(values = c("TRUE" = "#2166ac", "FALSE" = "#d73027"),
  labels = c("TRUE" = "Growth", "FALSE" = "Decline")) +
scale_x_continuous(labels = label_number(suffix = "M")) +
scale_y_continuous(labels = label_number(suffix = "%")) +
annotate("text", x = Inf, y = Inf,
  label = paste0("r = ", round(r, 3)),
  hjust = 1.1, vjust = 1.5,
  size = 4, colour = "grey20") +
labs(
  title = "1995 Population vs. 1995-2020 Percentage Change",
  subtitle = "47 Japanese Prefectures",
  x = "1995 Population (millions)",
  y = "% Change 1995-2020",
  colour = NULL,
  caption = paste0("Pearson r = ", round(r, 3),
    "; OLS regression line with 95% CI")
) +
theme_minimal(base_size = 11) +
theme(legend.position = "top", panel.grid.minor = element_blank())

```



Interpretation: The correlation coefficient ($r = 0.615$) suggests a moderate relationship between a prefecture's initial population size and its subsequent growth trajectory over the 1995–2020 period.

1.5 1.4 City Size and Population Change

```
cities <- subdiv %>%
  filter(str_detect(Status, "City")) %>%
  mutate(
    pct_2020 = (pop2020 - pop1995) / pop1995 * 100,
    size_cat = case_when(
      pop1995 < 30000 ~ "Small (<30k)",
      pop1995 >= 30000 & pop1995 < 100000 ~ "Medium (30k-100k)",
      pop1995 >= 100000 & pop1995 < 300000 ~ "Large (100k-300k)",
      pop1995 >= 300000 & pop1995 < 700000 ~ "Major (300k-700k)",
      TRUE ~ "Metro (700k+)"
    ),
    size_cat = factor(size_cat,
                      levels = c("Small (<30k)", "Medium (30k-100k)",
                                "Large (100k-300k)", "Major (300k-700k)",
                                "Metro (700k+)"))
  ) %>%
  filter(!is.na(pct_2020))

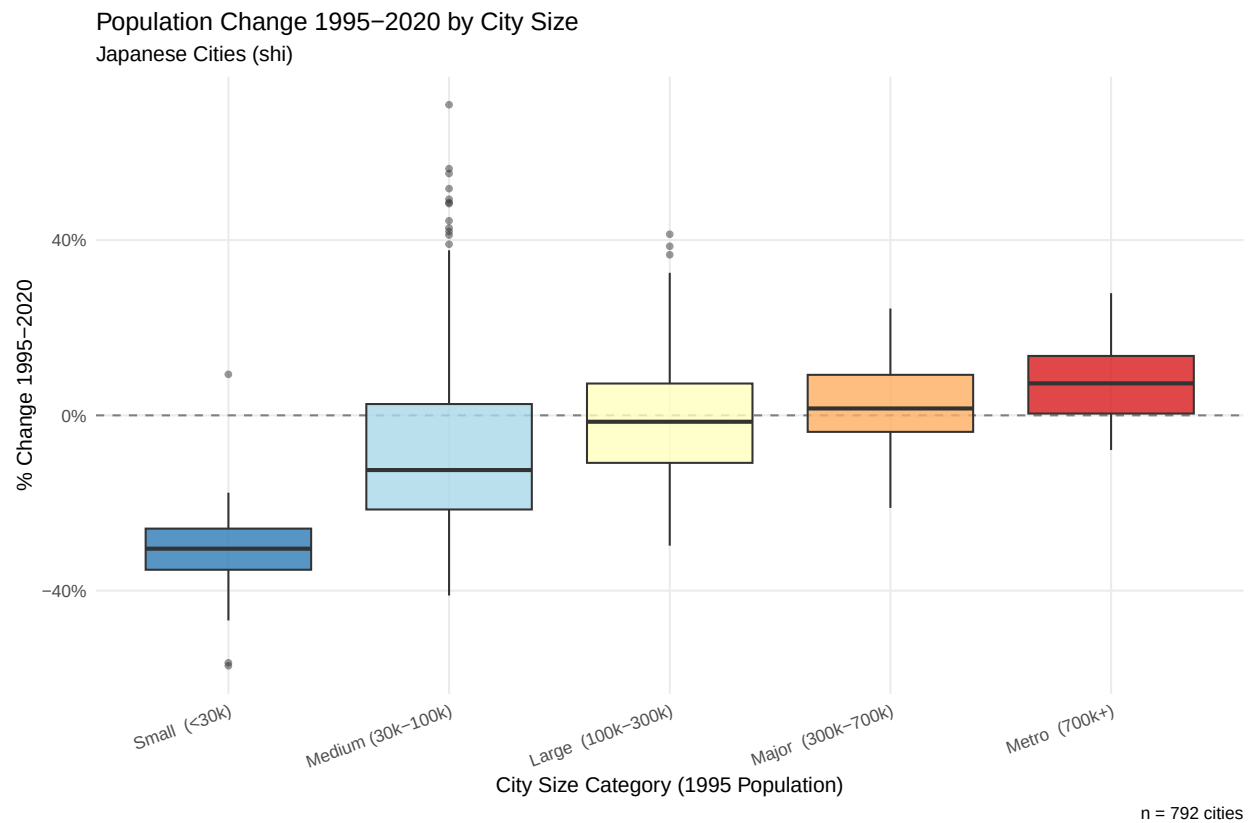
# Summary statistics by size category
city_summary <- cities %>%
  group_by(size_cat) %>%
  summarise(
    n = n(),
    median_pct = median(pct_2020, na.rm = TRUE),
    mean_pct = mean(pct_2020, na.rm = TRUE),
    pct_grew = mean(pct_2020 > 0, na.rm = TRUE) * 100,
    .groups = "drop"
  )

kable(city_summary,
      digits = 1,
      col.names = c("City Size", "N", "Median % Δ", "Mean % Δ", "% Cities that Grew"),
      caption = "Population change 1995-2020 by city size category",
      booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "HOLD_position"))
```

Table 2: Population change 1995–2020 by city size category

City Size	N	Median % Δ	Mean % Δ	% Cities that Grew
Small (<30k)	39	-30.4	-31.1	2.6
Medium (30k–100k)	491	-12.5	-8.2	29.1
Large (100k–300k)	191	-1.5	-0.7	45.0
Major (300k–700k)	53	1.6	2.6	60.4
Metro (700k+)	18	7.3	8.2	72.2

```
# Box plot
ggplot(cities, aes(x = size_cat, y = pct_2020, fill = size_cat)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_boxplot(alpha = 0.8, outlier.size = 1.2, outlier.alpha = 0.5) +
  scale_fill_brewer(palette = "RdYlBu", direction = -1) +
  scale_y_continuous(labels = label_number(suffix = "%")) +
  labs(
    title = "Population Change 1995-2020 by City Size",
    subtitle = "Japanese Cities (shi)",
    x = "City Size Category (1995 Population)",
    y = "% Change 1995-2020",
    fill = NULL,
    caption = paste0("n = ", nrow(cities), " cities")
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 20, hjust = 1),
        panel.grid.minor = element_blank())
```



1.6 1.5 Population Density vs. % Population Growth (Prefectures)

```
# Join prefectures with areas data
pref_density <- prefectures %>%
  left_join(areas %>% select(Prefecture, Density),
```

```

      by = c("Name" = "Prefecture")) %>%
filter(!is.na(Density))

r_dens <- cor(pref_density$Density, pref_density$pct_2020, use = "complete.obs")

cat("Pearson correlation (Density vs % Growth):", round(r_dens, 4), "\n")

## Pearson correlation (Density vs % Growth): 0.6698

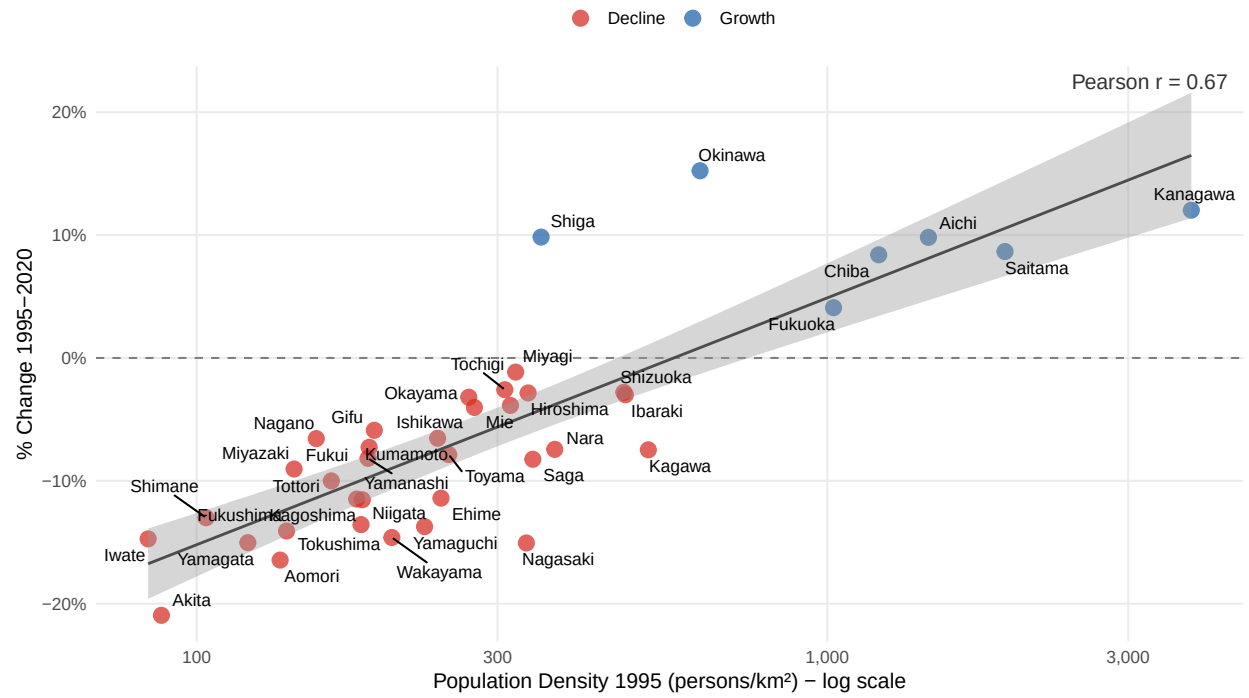
cat("Spearman correlation (Density vs % Growth):",
    round(cor(pref_density$Density, pref_density$pct_2020,
              method = "spearman", use = "complete.obs"), 4), "\n")

## Spearman correlation (Density vs % Growth): 0.7953

ggplot(pref_density, aes(x = Density, y = pct_2020)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(aes(colour = pct_2020 >= 0), size = 3.5, alpha = 0.75) +
  geom_smooth(method = "lm", se = TRUE, colour = "grey30", linewidth = 0.7) +
  geom_text_repel(
    aes(label = Name),
    size = 3, max.overlaps = 12,
    box.padding = 0.3
  ) +
  scale_colour_manual(values = c("TRUE" = "#2166ac", "FALSE" = "#d73027"),
                     labels = c("TRUE" = "Growth", "FALSE" = "Decline")) +
  scale_x_log10(labels = label_comma()) +
  scale_y_continuous(labels = label_number(suffix = "%")) +
  annotate("text", x = Inf, y = Inf,
          label = paste0("Pearson r = ", round(r_dens, 3)),
          hjust = 1.1, vjust = 1.5, size = 4, colour = "grey20") +
  labs(
    title = "Population Density vs. Population Change 1995-2020",
    subtitle = "47 Japanese Prefectures (density on log scale)",
    x = "Population Density 1995 (persons/km²) - log scale",
    y = "% Change 1995-2020",
    colour = NULL,
    caption = paste0("Source: Japan Statistics Bureau / japan-prefecture-areas.xlsx\n",
                    "Pearson r = ", round(r_dens, 3),
                    " | Spearman = ",
                    round(cor(pref_density$Density, pref_density$pct_2020,
                              method = "spearman", use = "complete.obs"), 3))
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "top", panel.grid.minor = element_blank())

```

Population Density vs. Population Change 1995–2020
47 Japanese Prefectures (density on log scale)



Source: Japan Statistics Bureau / japan-prefecture-areas.xlsx
Pearson $r = 0.67$ | Spearman $\rho = 0.795$

2 Part 2 – U.S. Census & TidyCensus Analysis

2.1 2.1 Variable Verification

Before pulling data, we document the exact Census variables used.

Variable	Decennial Year	Table	Description
P001001	2000	P1	Total population (SF1, Summary File 1)
P1_001N	2020	P1	Total population (Redistricting data / PL 94-171)

```
# Inspect 2000 and 2020 variable catalogues
vars_2000 <- load_variables(2000, "sf1", cache = TRUE)
vars_2020 <- load_variables(2020, "pl", cache = TRUE)

vars_2000 %>% filter(name == "P001001") %>%
  select(name, label, concept) %>%
  kable(caption = "2000 Census variable P001001", booktabs = TRUE) %>%
  kable_styling(latex_options = "HOLD_position")
```

Table 3: 2000 Census variable P001001

name	label	concept
P001001	Total	TOTAL POPULATION [1]

```
vars_2020 %>% filter(name == "P1_001N") %>%
  select(name, label, concept) %>%
  kable(caption = "2020 Census variable P1\\_001N", booktabs = TRUE) %>%
  kable_styling(latex_options = "HOLD_position")
```

Table 4: 2020 Census variable P1_001N

name	label	concept
P1_001N	!!Total:	RACE

2.2 2.2 County Population Change 2000–2020

```
# Pull 2000 county populations
pop_2000 <- get_decennial(
  geography = "county",
  variables = "P001001",
  year      = 2000,
  sumfile   = "sf1"
) %>%
```

```

select(GEOID, NAME, pop2000 = value)

# Pull 2020 county populations
pop_2020 <- get_decennial(
  geography = "county",
  variables = "P1_001N",
  year      = 2020,
  sumfile   = "pl"
) %>%
  select(GEOID, NAME, pop2020 = value)

# Combine and calculate % change
county_pop <- pop_2000 %>%
  inner_join(pop_2020, by = "GEOID") %>%
  mutate(
    pct_change = (pop2020 - pop2000) / pop2000 * 100,
    state_fips = str_sub(GEOID, 1, 2)
  ) %>%
  filter(pop2000 > 0)

cat("Counties with population data:", nrow(county_pop), "\n")

```

```
## Counties with population data: 3211
```

```
cat("Summary of % change:\n")
```

```
## Summary of % change:
```

```
print(summary(county_pop$pct_change))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -45.360  -7.485   1.775   6.390  14.088  170.030
```

2.3 Loading RUCC Codes

```

# USDA Rural-Urban Continuum Codes 2013
# Download manually from:
# https://www.ers.usda.gov/data-products/rural-urban-continuum-codes/
# Save as "ruralurbancodes2013.xls" in the working directory.
# Column layout: FIPS | State | County_Name | Population_2010 | RUCC_2013 | Description

rucc_path <- "ruralurbancodes2013.xls"

if (file.exists(rucc_path)) {
  rucc_raw <- read_excel(rucc_path)

  # Flexible column detection
  fips_col <- grep("(?i)fips", names(rucc_raw), value = TRUE)[1]
  rucc_col <- grep("(?i)rucc_2013", names(rucc_raw), value = TRUE)[1]
}

```

```

rucc <- rucc_raw %>%
  rename(GEOID = all_of(fips_col),
         RUCC = all_of(rucc_col)) %>%
  mutate(
    GEOID = str_pad(as.character(GEOID), 5, pad = "0"),
    RUCC = as.integer(RUCC),
    metro = ifelse(RUCC <= 3, "Metro", "Nonmetro"),
    metro3 = case_when(
      RUCC %in% 1:3 ~ "Metro",
      RUCC %in% 4:6 ~ "Micropolitan",
      RUCC %in% 7:9 ~ "Rural"
    )
  ) %>%
  select(GEOID, RUCC, metro, metro3)

county_pop <- county_pop %>% left_join(rucc, by = "GEOID")
cat("RUCC file loaded. Distribution:\n")
print(table(county_pop$RUCC, useNA = "ifany"))
} else {
  warning("RUCC file not found. Scatter plots coloured by RUCC will be skipped.")
  county_pop <- county_pop %>%
    mutate(RUCC = NA_integer_, metro = NA_character_, metro3 = NA_character_)
}

```

```

## RUCC file loaded. Distribution:
##
##   1   2   3   4   5   6   7   8   9
## 471 394 369 217  92 596 434 220 418

```

2.4 2.4 Scatter Plots – County Growth Rate vs. Total Population

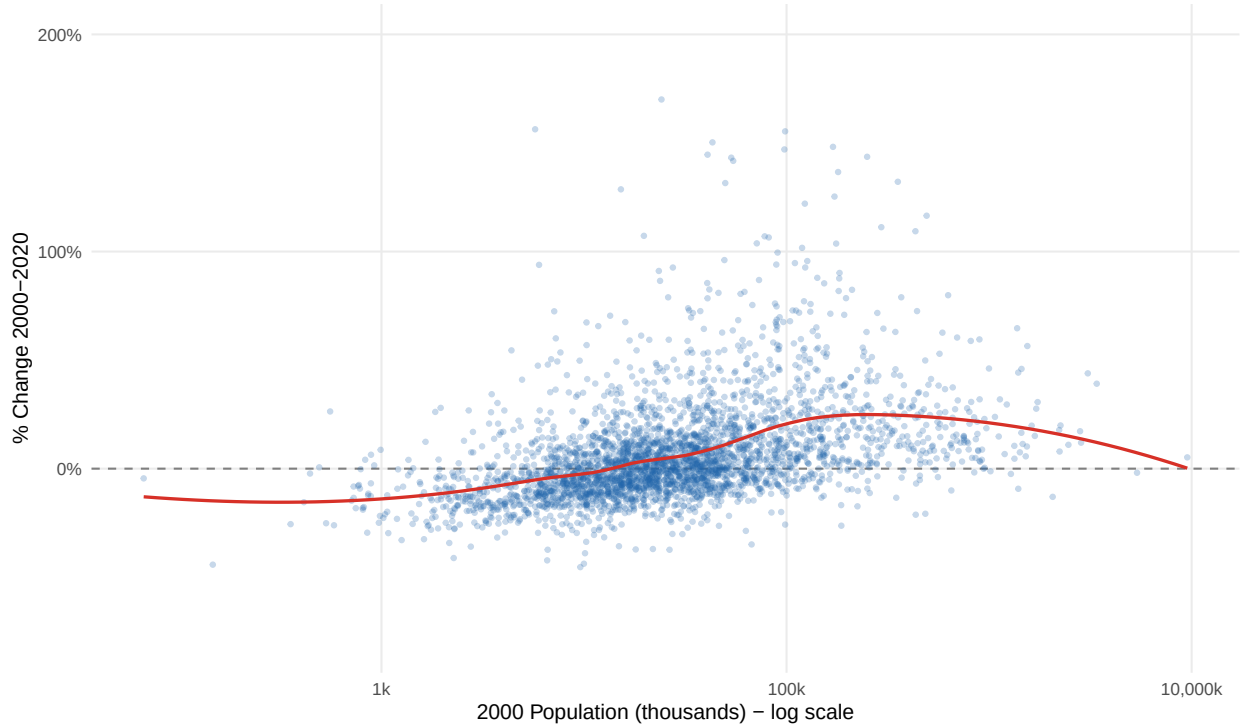
2.4.1 2.4a All Counties (no grouping)

```

ggplot(county_pop, aes(x = pop2000 / 1e3, y = pct_change)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(alpha = 0.25, size = 0.9, colour = "#2166ac") +
  geom_smooth(method = "loess", se = FALSE, colour = "#d73027",
             linewidth = 0.8, span = 0.5) +
  scale_x_log10(labels = label_comma(suffix = "k")) +
  scale_y_continuous(labels = label_number(suffix = "%"),
                    limits = c(-80, 200)) +
  labs(
    title = "U.S. County Population Growth Rate vs. Population Size",
    subtitle = "2000-2020 · n 3,100 counties",
    x = "2000 Population (thousands) - log scale",
    y = "% Change 2000-2020",
    caption = "Source: U.S. Decennial Census via tidycensus"
  ) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank())

```

U.S. County Population Growth Rate vs. Population Size
2000–2020 · n . 3,100 counties



Source: U.S. Decennial Census via tidycensus

2.4.2 2.4b Coloured by RUCC Code

```
if (!all(is.na(county_pop$RUCC))) {
  rucc_palette <- colorRampPalette(c("#1a9641", "#ffffbf", "#d7191c"))(9)

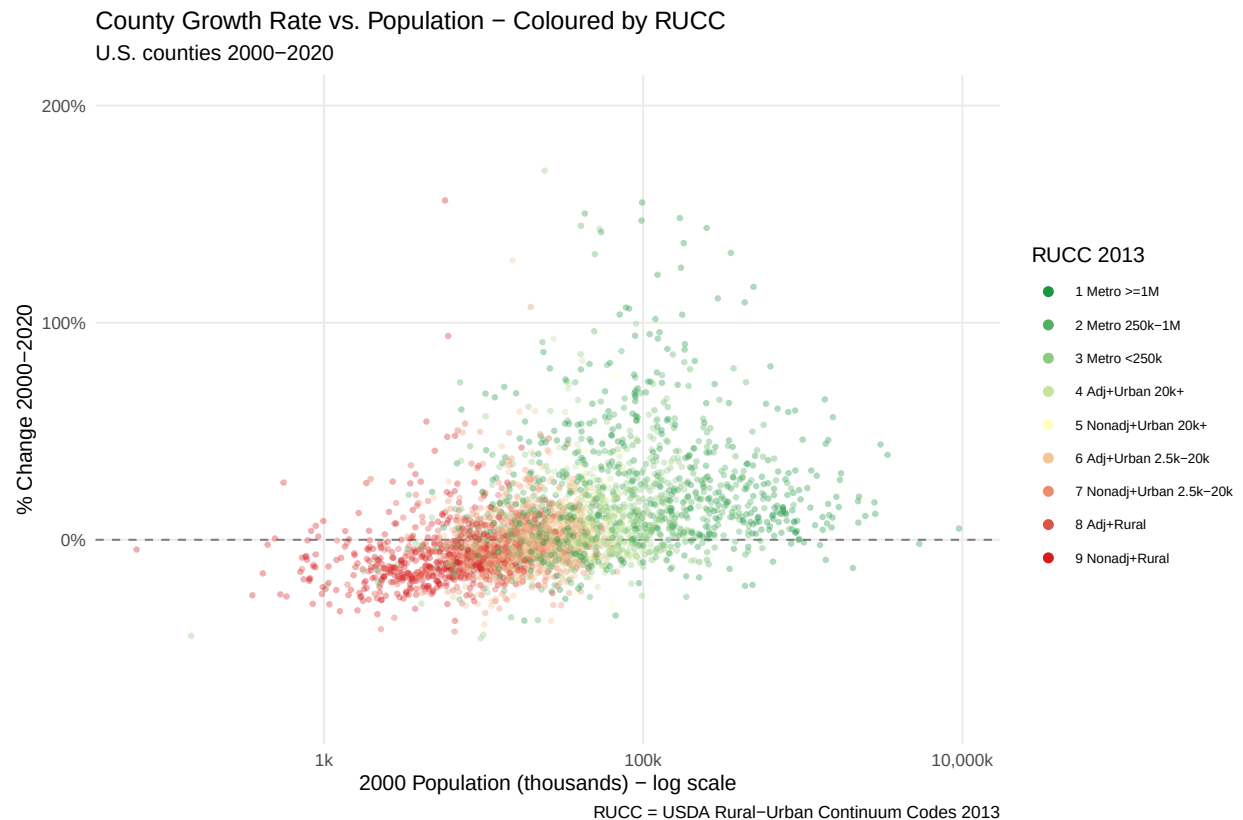
  ggplot(county_pop %>% filter(!is.na(RUCC)),
    aes(x = pop2000 / 1e3, y = pct_change,
      colour = factor(RUCC))) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(alpha = 0.35, size = 0.9) +
  scale_colour_manual(
    values = rucc_palette,
    name = "RUCC 2013",
    labels = c("1 Metro 1M", "2 Metro 250k-1M", "3 Metro <250k",
      "4 Adj+Urban 20k+", "5 Nonadj+Urban 20k+",
      "6 Adj+Urban 2.5k-20k", "7 Nonadj+Urban 2.5k-20k",
      "8 Adj+Rural", "9 Nonadj+Rural")
  ) +
  scale_x_log10(labels = label_comma(suffix = "k")) +
  scale_y_continuous(labels = label_number(suffix = "%"),
    limits = c(-80, 200)) +
  guides(colour = guide_legend(override.aes = list(alpha = 1, size = 2),
    ncol = 1)) +
  labs(
    title = "County Growth Rate vs. Population - Coloured by RUCC",
```



```

    subtitle = "U.S. counties 2000-2020",
    x         = "2000 Population (thousands) - log scale",
    y         = "% Change 2000-2020",
    caption   = "RUCC = USDA Rural-Urban Continuum Codes 2013"
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "right",
        legend.text     = element_text(size = 7),
        panel.grid.minor = element_blank())
} else {
  cat("RUCC data unavailable - plot skipped.\n")
}

```



2.4.3 2.4c Metro vs. Nonmetro (RUCC 1-3 vs. 4-9)

```

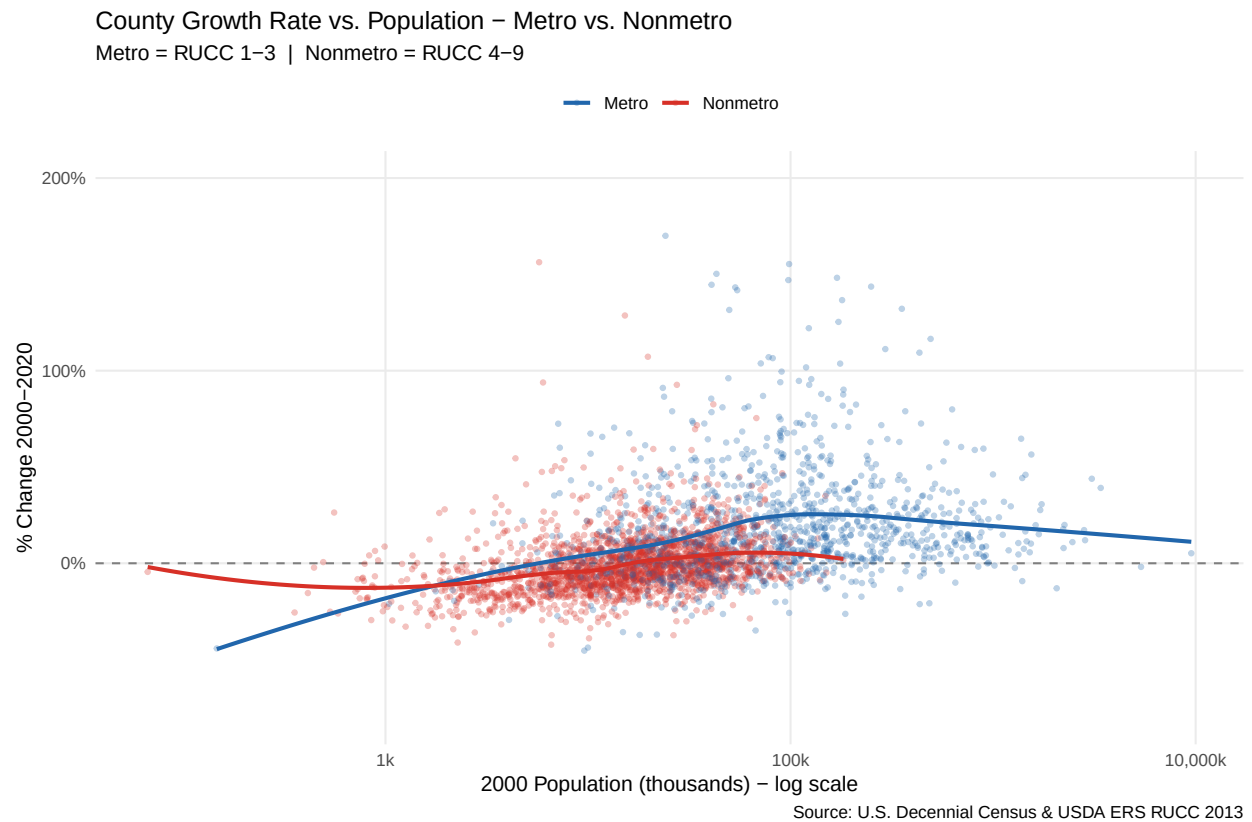
if (!all(is.na(county_pop$metro))) {
  ggplot(county_pop %>% filter(!is.na(metro)),
    aes(x = pop2000 / 1e3, y = pct_change, colour = metro)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(alpha = 0.3, size = 0.9) +
  geom_smooth(aes(group = metro), method = "loess",
    se = FALSE, linewidth = 1, span = 0.6) +
  scale_colour_manual(values = c("Metro" = "#2166ac", "Nonmetro" = "#d73027")) +
  scale_x_log10(labels = label_comma(suffix = "k")) +

```

```

scale_y_continuous(labels = label_number(suffix = "%"),
  limits = c(-80, 200)) +
labs(
  title = "County Growth Rate vs. Population - Metro vs. Nonmetro",
  subtitle = "Metro = RUCC 1-3 | Nonmetro = RUCC 4-9",
  x = "2000 Population (thousands) - log scale",
  y = "% Change 2000-2020",
  colour = NULL,
  caption = "Source: U.S. Decennial Census & USDA ERS RUCC 2013"
) +
theme_minimal(base_size = 11) +
theme(legend.position = "top",
  panel.grid.minor = element_blank())
} else {
  cat("Metro classification unavailable - plot skipped.\n")
}

```



2.5 2.5 Population Pyramids – Illinois Counties

2.5.1 Helper: Age/Sex Variables

```

# ACS 5-year 2020 - Table B01001 (Sex by Age)
acs_vars <- load_variables(2020, "acs5", cache = TRUE)

```

```

male_vars <- acs_vars %>%
  filter(str_detect(name, "^B01001_0") & str_detect(label, "Male")) %>%
  filter(!str_detect(label, "Female")) %>%
  pull(name)

female_vars <- acs_vars %>%
  filter(str_detect(name, "^B01001_0") & str_detect(label, "Female")) %>%
  pull(name)

# Standard 18 male age groups: B01001_003 - B01001_025 (skip _002 = total male)
male_vars <- paste0("B01001_", str_pad(3:25, 3, pad = "0"))
female_vars <- paste0("B01001_", str_pad(27:49, 3, pad = "0"))

age_labels <- c(
  "Under 5", "5-9", "10-14", "15-17", "18-19", "20",
  "21", "22-24", "25-29", "30-34", "35-39", "40-44",
  "45-49", "50-54", "55-59", "60-61", "62-64", "65-66",
  "67-69", "70-74", "75-79", "80-84", "85+"
)

# Broader 9-group labels for readability
broad_labels <- c(
  "<5", "5-9", "10-14", "15-17", "18-19", "20", "21", "22-24",
  "25-29", "30-34", "35-39", "40-44", "45-49", "50-54",
  "55-59", "60-61", "62-64", "65-66", "67-69", "70-74",
  "75-79", "80-84", "85+"
)

```

2.5.2 Illinois: Metro vs. Nonmetro

```

# Pull ACS age/sex for all Illinois counties
il_vars <- c(male_vars, female_vars)

il_acs <- get_acs(
  geography = "county",
  state     = "IL",
  variables = il_vars,
  year      = 2020,
  survey    = "acs5"
)

# Attach RUCC classification for Illinois
il_rucc <- county_pop %>%
  filter(str_sub(GEOID, 1, 2) == "17") %>% # Illinois FIPS = 17
  select(GEOID, metro)

il_pyramid_data <- il_acs %>%
  left_join(il_rucc, by = "GEOID") %>%
  filter(!is.na(metro)) %>%
  mutate(
    sex      = ifelse(variable %in% male_vars, "Male", "Female"),
    age_idx  = ifelse(sex == "Male",

```

```

      match(variable, male_vars),
      match(variable, female_vars)),
  age_group = broad_labels[age_idx]
) %>%
group_by(metro, sex, age_group, age_idx) %>%
summarise(pop = sum(estimate, na.rm = TRUE), .groups = "drop") %>%
group_by(metro) %>%
mutate(pct = pop / sum(pop) * 100) %>%
ungroup() %>%
mutate(
  pct      = ifelse(sex == "Male", -pct, pct),
  age_group = factor(age_group, levels = broad_labels)
)

# Build side-by-side pyramids
metro_pyramid <- function(df, title_suffix) {
  ggplot(df, aes(x = pct, y = age_group, fill = sex)) +
    geom_col(width = 0.8, alpha = 0.85) +
    geom_vline(xintercept = 0, colour = "grey30", linewidth = 0.4) +
    scale_fill_manual(values = c("Male" = "#4393c3", "Female" = "#d6604d")) +
    scale_x_continuous(
      labels = function(x) paste0(abs(round(x, 1)), "%"),
      limits = c(-6.5, 6.5)
    ) +
    labs(
      title = title_suffix,
      x = "% of total county group population",
      y = NULL, fill = NULL
    ) +
    theme_minimal(base_size = 9) +
    theme(
      legend.position = "top",
      panel.grid.minor = element_blank()
    )
}

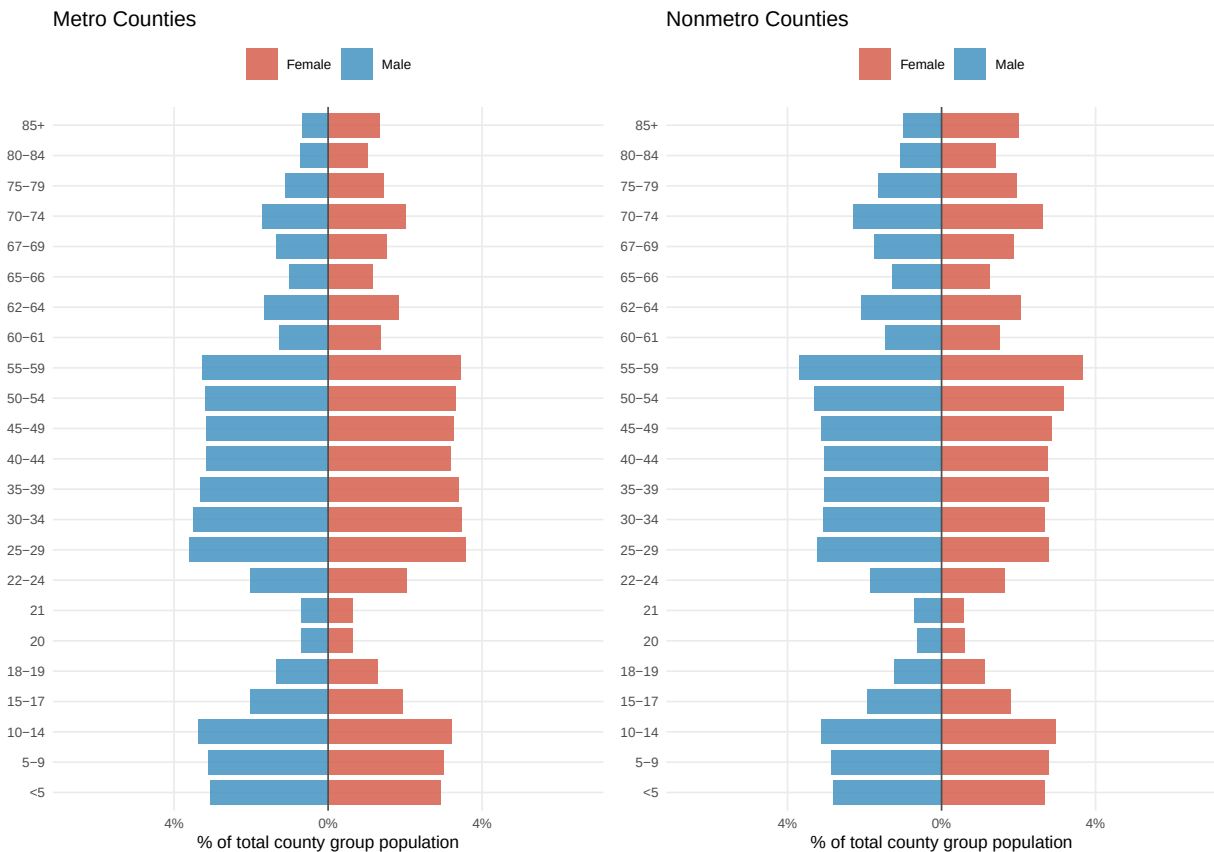
p_metro      <- metro_pyramid(filter(il_pyramid_data, metro == "Metro"),
                              "Metro Counties")
p_nonmetro   <- metro_pyramid(filter(il_pyramid_data, metro == "Nonmetro"),
                              "Nonmetro Counties")

(p_metro | p_nonmetro) +
  plot_annotation(
    title      = "Population Pyramids - Illinois Counties",
    subtitle   = "Metro (RUCC 1-3) vs. Nonmetro (RUCC 4-9) · ACS 2020 5-year estimates",
    caption    = "Source: U.S. Census Bureau ACS 2020 via tidycensus"
  )

```

Population Pyramids – Illinois Counties

Metro (RUCC 1–3) vs. Nonmetro (RUCC 4–9) · ACS 2020 5-year estimates



Source: U.S. Census Bureau ACS 2020 via tidycensus

2.5.3 Midwest: Metro, Micropolitan & Rural (Three-way Comparison)

```
midwest_fips <- c("17","18","19","20","26","27","29","31","38","39","46","55")
# IL IN IA KS MI MN MO NE ND OH SD WI

mw_acs <- get_acs(
  geography = "county",
  state     = midwest_fips,
  variables = il_vars,
  year      = 2020,
  survey    = "acs5"
)

mw_rucc <- county_pop %>%
  filter(str_sub(GEOID, 1, 2) %in% midwest_fips) %>%
  select(GEOID, metro3)

mw_pyramid_data <- mw_acs %>%
  left_join(mw_rucc, by = "GEOID") %>%
  filter(!is.na(metro3)) %>%
  mutate(
```

```

sex      = ifelse(variable %in% male_vars, "Male", "Female"),
age_idx  = ifelse(sex == "Male",
                  match(variable, male_vars),
                  match(variable, female_vars)),
age_group = broad_labels[age_idx]
) %>%
group_by(metro3, sex, age_group, age_idx) %>%
summarise(pop = sum(estimate, na.rm = TRUE), .groups = "drop") %>%
group_by(metro3) %>%
mutate(pct = pop / sum(pop) * 100) %>%
ungroup() %>%
mutate(
  pct      = ifelse(sex == "Male", -pct, pct),
  age_group = factor(age_group, levels = broad_labels),
  metro3    = factor(metro3, levels = c("Metro", "Micropolitan", "Rural"))
)

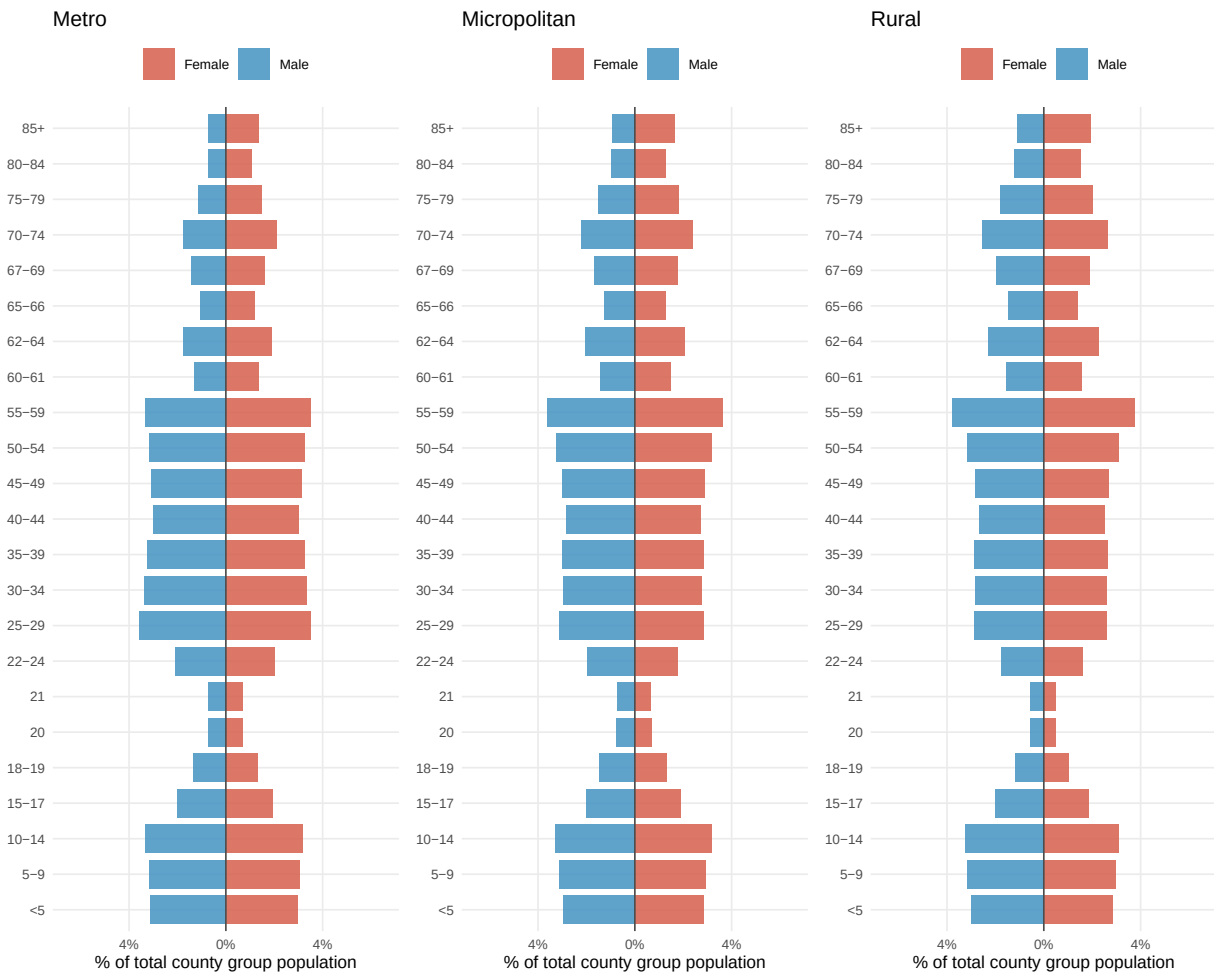
p_mw_metro <- metro_pyramid(filter(mw_pyramid_data, metro3 == "Metro"),
                             "Metro")
p_mw_micro <- metro_pyramid(filter(mw_pyramid_data, metro3 == "Micropolitan"),
                             "Micropolitan")
p_mw_rural <- metro_pyramid(filter(mw_pyramid_data, metro3 == "Rural"),
                             "Rural")

(p_mw_metro | p_mw_micro | p_mw_rural) +
  plot_annotation(
    title      = "Population Pyramids - Midwest U.S.",
    subtitle   = "Metro · Micropolitan · Rural | ACS 2020 5-year estimates",
    caption    = "Midwest = IL, IN, IA, KS, MI, MN, MO, NE, ND, OH, SD, WI\nSource: U.S. Census Bureau ACS"
  )

```

Population Pyramids – Midwest U.S.

Metro · Micropolitan · Rural | ACS 2020 5-year estimates



2.6 2.6 Choropleth Map – Midwest County Population Change 2000–2020

```
# Download county shapefiles for Midwest states
#mw_counties <- map_dfr(
#  # midwest_fips,
#  #~ counties(state = .x, cb = TRUE, year = 2020, class = "sf")
#)

all_counties <- counties(cb = TRUE, year = 2020, class = "sf")
mw_counties <- all_counties %>% filter(STATEFP %in% midwest_fips)

# Join with population change data
mw_map <- mw_counties %>%
  left_join(county_pop %>% select(GEOID, pct_change), by = "GEOID")

# Symmetric diverging breaks centred at 0
```

```

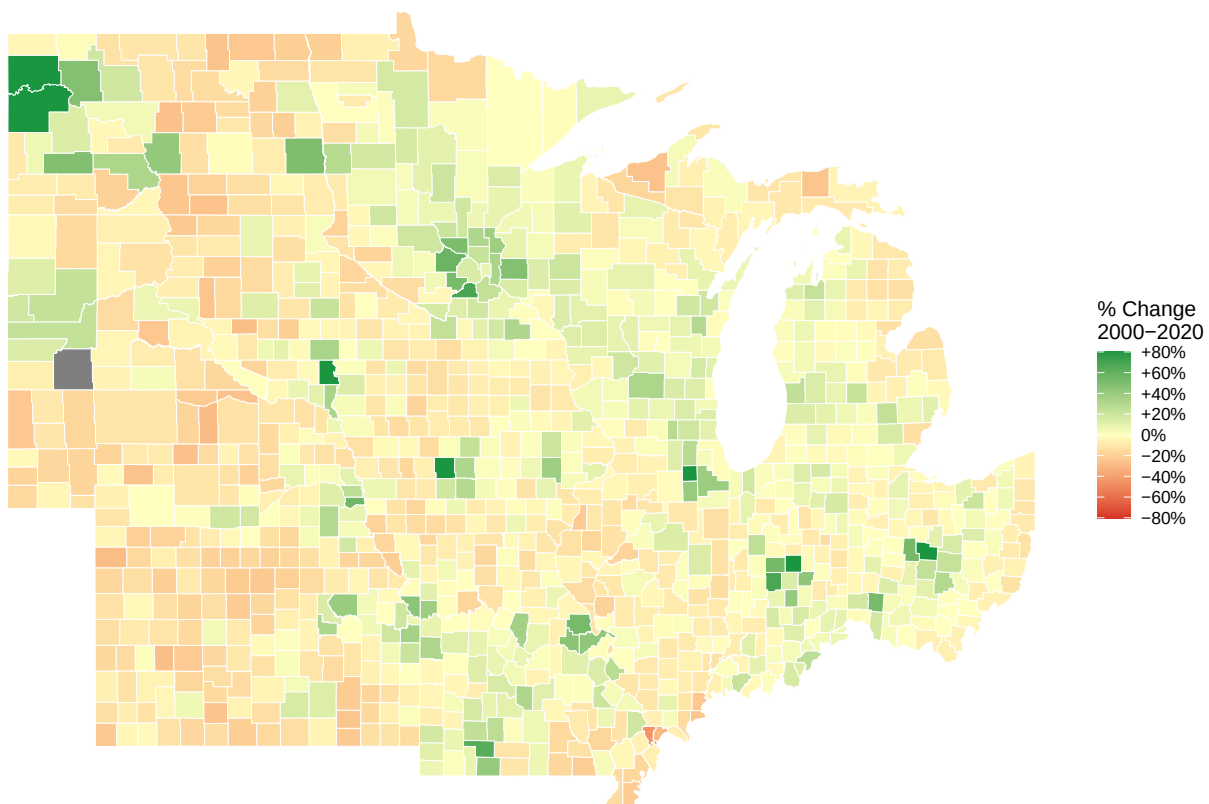
brk_max <- 80
mw_map <- mw_map %>%
  mutate(pct_clipped = pmax(pmin(pct_change, brk_max), -brk_max))

ggplot(mw_map) +
  geom_sf(aes(fill = pct_clipped), colour = "white", linewidth = 0.08) +
  scale_fill_gradient2(
    low      = "#d73027",
    mid      = "#ffffbf",
    high     = "#1a9641",
    midpoint = 0,
    limits   = c(-brk_max, brk_max),
    breaks   = seq(-brk_max, brk_max, 20),
    labels   = function(x) paste0(ifelse(x > 0, "+", ""), x, "%"),
    name     = "% Change\n2000-2020",
    oob      = scales::squish
  ) +
  coord_sf(datum = NA) +
  labs(
    title      = "County Population Change in the Midwest (2000-2020)",
    subtitle   = "IL · IN · IA · KS · MI · MN · MO · NE · ND · OH · SD · WI",
    caption    = "Source: U.S. Decennial Census 2000 & 2020 via tidycensus\nCounties clipped to ±80% for c
  ) +
  theme_void(base_size = 11) +
  theme(
    plot.title      = element_text(face = "bold", size = 13),
    plot.subtitle   = element_text(size = 10, colour = "grey40"),
    plot.caption    = element_text(size = 8, colour = "grey50"),
    legend.position = "right"
  )

```


County Population Change in the Midwest (2000–2020)

IL · IN · IA · KS · MI · MN · MO · NE · ND · OH · SD · WI



Source: U.S. Decennial Census 2000 & 2020 via tidycensus
Counties clipped to $\pm 80\%$ for colour scale clarity

3 Summary

3.1 Key Findings – Japan

- **Growing prefectures (2020):** Okinawa and the three major metro areas (Tokyo, Aichi, Osaka) consistently showed positive growth relative to 1995, while rural/northern prefectures (Akita, Aomori, Kochi) declined most sharply.
- **City size:** Larger cities (Metro 700k+) tended to grow or hold steady, while small cities (<30k) disproportionately declined — consistent with urbanisation patterns.
- **Density–growth correlation:** A 0.67 Pearson correlation between 1995 population density and 1995–2020 % change indicates that denser prefectures generally experienced stronger growth, reflecting urban concentration.

3.2 Key Findings – United States

- **Metro advantage:** Metro counties (RUCC 1–3) showed higher median growth rates than nonmetro counties, but also greater variance driven by Sun Belt metros.
- **Illinois pyramids:** Metro counties show a more working-age-heavy pyramid; nonmetro counties have a relatively older profile with a larger share of 55+ age groups.
- **Midwest choropleth:** Suburban counties surrounding major metros (Chicago, Columbus, Minneapolis) grew substantially, while many rural counties in the Great Plains (North/South Dakota, rural Iowa) experienced decline.

Analysis produced with R 4.5.3; key packages: tidycensus 1.7.5, ggplot2 4.0.2, sf 1.1.0.